Merging and Splitting Maneuver of Platoons by Means of a Novel PID Controller

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***Abstract*— With the advent of self-driving cars in road networks, automobile engineers have now focused on platooning of Heavy Duty Vehicles (HDVs) to look into the freight transport domain. A platoon is a number of vehicles travelling in tandem and electronically connected to each other. In a platoon, the first vehicle has a certified driver and the other vehicles follow autonomously. While platooning reduces the number of drivers generally required for standalone vehicles, it also improves fuel efficiency reducing the air-drag between the trails of vehicles. Although a number of researchers in the world have concentrated on platooning from the application point of view, this paper proposes a control algorithm which looks into the basic platooning maneuvers, such as platoon merging and platoon splitting. The merging and splitting of platoons in this work has been realized using the most commonly used controller used in the process industry, i.e. the PID controller which has been simulated using the micro-simulator VISSIM. The results presented in the end of the paper shows the efficacy of the PID controller used for platoon movements as well as for platoon maneuvers as well.**

Keywords*— *PID controller, Platooning, Platoon Splitting, Platoon Merging, Constant distance topology, VISSIM, MATLAB

I. INTRODUCTION

The increasing dependence on transportation in the recent days have eased our lives in many ways. The continuous accruing need of this dependence has led to various problems in terms of traffic congestion and increased volume of emitted harmful compounds. In the past years, there has been an exponential increase in the demand for road based freight transportation. The major reprise to such an increased demand is the concept of heavy duty platooning. Platooning can be seen as the concept of railways transcended to a highway road networks between cities as well as across countries. Platooning enables a number of vehicles to drive within a short, acceptable distance from each other thereby allowing them to act together as one unit. With vehicles moving in a platoon, there has always been a steady increase in the traffic flow with reduced human labor. While on one hand, where the platoons travelling as a single unit can save a lot of space on freeways to accommodate much more traffic, on the other hand they also reduce fuel consumption by significant reduction in air drag. Statistically freight transport in the world consumes million tons of fuel every year and an even modest decrease in fuel consumption can lead to humongous savings for thousands of vehicles on a much broader scale.

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Although the study of platoons has been in vogue for a number of years now [1], most of the methodologies have concentrated on the platoon control of vehicles using the well-known strategies such as the Adaptive Cruise Control (ACC) or its most recent and sophisticated version i.e., Cooperative Adaptive Cruise Control (CACC). Rajamani *et.al.* in [2] have implemented platooning and considered it as an extension to a simple Cooperative Control (CC) strategy by introducing the braking system for the followers to imitate. While the needs of intermediate end users considering their safety and the operational benefits have been considered by the project PROMOTE-CHAUFFEUR I and II [3], the KONVOI [4] project firmly looks into the movement of electronically operated trucks in a 5 vehicle platoon. One of the most talked-about research projects under the California Institute of Technology is the PATH project which deals with multiple issues of platooning and mainly addresses the traffic related issues [5]. The ENERGY ITS [6] project evaluated the energy efficiency for automated platoons and concentrated on energy savings which is one of the prime deliverables of recent day research. Another talked about research project has been the SARTRE project [7]. The crux of the same had been the analysis of mixed traffic in highway situations considering issues of safety, comfort and fuel efficiency. While the most recent project on platooning, i.e., the COMPANION project [8] focus on the creation, operation and coordination of platoons, most of the literature above broadly overlooks platooning from the application point of view. These papers use the generic CACC algorithm, which the work in this paper choses to improve and apply for platoon maneuvers such as splitting and merging. Vehicle platooning has been extensively dealt by Alam et.al [9] [10] [11] although most of their research focus primarily on fuel usage reduction. It is a well-known concept that for heavy duty platooning, if the platoon members move with the least possible distance, the air-drag between the vehicles decrease improving the fuel efficiency [11]. This has been the nub of most of the literature in [9] [11] [12]. The use of optimal control in platooning has been considered by a number of researchers [19] and so has been dealt by Alam in [13]. In [13], Alam.et.al designed a system model for a Heavy Duty Platoon and determined the fuel efficient behavior of the same while traversing an up-hill or a down-hill terrain on Swedish Freeways. The primary reason being that LQR doesn't have a proper solution all the time as the performance of the same depends under the inherent supposition that the desired final state is reachable from the given initial state.

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Incase this condition fails, it is impossible to build an input that satisfies the primary requirement for the solution to exist. Even if the LQR is solved, there is a small chance that the resulting system would be stable with acceptable closed loop specifications of the system. Further the standard LQR design has no inhibitions on the input signal which is impractical for real-time systems such as vehicular platooning which can be explained with a simple example. For platoons moving in inclined highways, the LQR solution can provide an unbounded input torque that may be required to propel the vehicle forward but in actual practice it can prove detrimental for the vehicle mechanisms. Optimizing a system performance on the basis of a single criterion such as decreasing the cost-function can sometimes affect the other overall system performances as well as other system parameters. Hence LQR technique is not a pragmatic approach for real time used cases as HDV platooning.

Vehicular platooning using CACC enabled by VANET has been shown by Amoozadeh *et.al.* in [18]. In this work, the two layer controllers are developed- i.e. the upper layer and the lower layer. While the upper layer controller has certain modes of operation such as the speed control mode (SC), gap control mode (GC) and the collision avoidance mode (CA), the lower controller determines the throttle and/or brake commands to track the desired acceleration of the vehicles moving in the platoon. While the platoons are in operation, they need to shift between these modes, i.e. the SC mode, the GC mode and the CA mode depending on inter-vehicular distances and the platoon maneuvers like merging and splitting. Vehicles moving at high speeds together with a communication frequency of 10 Hz and with a minimum gap of 2 meters would have a higher risk of collision because of the shift in above controller modes which that might take up the crucial seconds needed once an emergency situation arise. The comparison of the controller strategy used in [18] has also been shown later in this paper.

The rest of the paper is organized as follows: Section II deals with the modeling of the vehicular platoon system considering number of trucks in the platoon. The nomenclature of the header, the leader and the follower vehicle has been clearly indicated with their implications that they have on the controller performance has been mentioned in this section. Section III deals with the platoon maneuvers and how the splitting and merging has been realized in the Singapore road network. Section IV shows the simulation results and curves of the speeds and intervehicular distance of the platoons while the splitting and merging maneuvers happen. The comparison of this methodology with [18] has been elucidated in section V using speed and distance curves and finally, Section VI deals with the conclusion and the future scope of work.

II. Platoon Nomenclature and Modelling

This section of the paper deals with platoon modelling terminologies and the controller developed that has been used for the platoon maneuver operations.

a) ***Introduction***: One of the major topics of interest for automobile engineers has been the concept of vehicle moving in tandem in either longitudinal or lateral fashion. This is of high interest as it reduces traffic congestion, fuel consumption, and improves driver comfort along with limiting heavy emissions. Another major advantage of platooning circumscribes the concept of the movement of vehicles together in a unison which transcends the concepts of railways to roads adding up to the advantage of higher flexibility of freight transport in terms of reaching any remote location. The main attribute of vehicular platooning has been to maintain a fixed speed that imitates the leader as closely as possible and to have a pre-described fixed distance (**$GAP_{DESIRED}$**) between the members of the platoon. There are two major kinds of topologies used, namely the constant distance topology and the headway time topology. In the constant distance topology, the vehicles would try to maintain the same intervehicular distance irrespective of the vehicle's velocity profile which differs from the headway time topology. In the headway time topology, the distance between the vehicles are a function of their individual velocities. The vehicles travelling at higher speed would hence require much more safety distances than the one travelling at slower speeds. The work done in this paper focusses mainly on the first protocol of operation, i.e. the constant distance topology where the vehicles would be moving in unison keeping a fixed inter-vehicular distance, i.e. **$GAP_{DESIRED}$** between them. The basic advantage of using this topology has been the factor of safety which can be elucidated as follows;

Let us consider a platoon moving within a congested area. While the speed of the platoon is low, the distances between them is very small as per the headway time protocol. In such cases, if there has been consecutive data drops, the chances of collision exponentially increases. This issue can be tackled using the constant distance topology as the distance would be same at any speeds. The time-headway policy has another major disadvantage; while the vehicles are moving at high speeds as in highways, the distance between the vehicles would increase and let other non-platoon vehicles to come in between the platoon leading to communication drops between the members of the platoon. This constant distance methodologies takes into account of this drop and let the control designer to plan the platoon with a pre-fixed distance between them even taking account of the different length of trucks of different makes.

b) ***Terminology***: This section depicts the terminology of how the platoon of vehicles have been articulated for the paper and has been used in the rest of the paper henceforth.

Figure 1: A vehicle Platoon for 3 HDVs.

Figure 1 above shows a case of 3 HDVs travelling in unison, depicting their direction of motion with wireless communication between them. For this case, the first vehicle would be termed as a *header*, which would be same once the platoon extends for a n number of vehicles. The header vehicle is driven by a certified driver who determines the speed of platoon and hence it is imitated by the other vehicles of the platoon autonomously. It can be seen that for the second vehicle, the first vehicle would be considered as the header as well as the leader, while for the third vehicle, the second one would be considered as the leader. The first one would always be the header in any case. Similarly extending this concept for a representative case of n member platoon, it can be written that the n^{th} vehicle of a platoon would be the follower while the $(n-1)^{th}$ vehicle would be leader.

c) ***Modelling of the Platoon and the Controller:*** This section of the paper focuses on the mathematical modeling of the platoon for longitudinal motion of vehicles. A vehicle platooning system can be considered similar to a mass-spring-damper system [15]. Similar to the mass spring damper system, the platoon of vehicles follow the control law of

$$m_i\ddot{x}_i + b_i\dot{x}_i = u_i \tag{1}$$

where the index i denotes the i^{th} vehicle in the platoon, m_i is the mass, b_i is the effect of damping, x_i is the longitudinal position of the i^{th} vehicle and u_i is the input information of the same. In this work, the controller has been designed on the basis of the most well-known control algorithm used in process industries, i.e. the Proportional Integral Derivate (PID) controller. The main aim of the controller for this scenario would be to stabilize a platoon of length n, such that the follower would be tracking the leader vehicle in a longitudinal fashion. The crux of the control would be to keep a safe inter-vehicular distance and to maintain a relatively constant speed with the least use of brakes while in motion and applying them as fast as possible in case of an emergency. Ideally in such cases, the step response of the platooning system would normally be an over-damped one. The primary idea behind this has been the fact that if the system is under-damped, it would take more time for the followers to react to the control commands fed to it and might lead to catastrophes while the vehicles are travelling at high speeds. It has also to be borne in mind that the rise time and the settling time of the response should be as less as possible so that while the vehicles are in motion, the platoon operates as a single unit and that is why PID controllers have been chosen for this work. PID controllers are also the most easily implementable ones. The PID gains are designed based on the tracking error and all the gains of the system are balanced to attain a trade-off between the acceptable system responses mentioned above.

A conventional PID controller is formulated as

$$u(t) = k_p e(t) + k_i \int_0^t e(t)dt + k_d \frac{d}{dt}e(t) \tag{2}$$

where k_p, k_i and k_d represents the proportional, integral and derivative gain of the system respectively. The corresponding transfer function $G(s)$ from the error signal $e(t)$ to the input signal can be written as:

$$G(s) = \frac{U(S)}{E(S)} = k_p + \frac{k_i}{s} + k_d s = \frac{k_d s^2 + k_p s + k_i}{s} \tag{3}$$

For this control problem, the difference between positions of the leader and the follower has been taken as an error signal for the controller. This is a very pragmatic approach for longitudinal control of vehicle platooning. The controller receives the reference position x_{ref} which is the position of the leader x_L and the position of the follower x_F for each of the i vehicles. The difference of the same is fed to the controller as error signal and the velocity of the system has been calculated at every instance based on the acceleration. The acceleration of the follower vehicle has been calculated for each leader-follower set of vehicles.

Using the concepts mentioned in [7], the force required for moving a platoon of vehicle can be considered as

$$m\dot{u} = F_x - mg\sin\theta - f_r mg\cos\theta - \frac{1}{2}C_{air}(u+u_w)^2 \tag{4}$$

where the values of parameters and their definitions have been illustrated in Table 1.

Table 1 : Parameters and their values

Parameter Symbols	Definition
F_x	Tractive Force
C_d	Drag Coefficient
A_f	Frontal Area of the Vehicle
u_w	Wind Velocity
θ	Angle of Inclination
u	Vehicle Velocity
u_w	Wind velocity
C_{air}	$\rho A_r C_d$
m	Mass of the Vehicle
g	Acceleration Due to Gravity
f_r	Rolling Resistance Co-efficient

This is to be noted that u_w is the wind velocity which is positive for a headwind and negative in-case of a tailwind. The drag co-efficient C_d ranges from 0.2 to 1.5 for HDVs.

For formulating the PID controller, few variables have been assigned. The variable dx_{fl} represents the distance between the $(i-1)^{th}$ & the i^{th} vehicle of the platoon, where i^{th} is the corresponding follower vehicle in the platoon whose acceleration is to be determined. x_l, x_f & x_h represents the coordinate positions of the leader, follower and header respectively. The velocity and acceleration of follower vehicle are represented by v_f & a_f . v_h & a_h signifies the velocity and acceleration of header vehicle. Considering dx_{fh} as the distance between the header & the i^{th} vehicle in the platoon, the PID controller in (3) can be written as

$$a_{f_i} = \left[\frac{(a_h + a_l)k_d + (v_h - v_f)k_p + (v_l - v_f)k_p + k_i(x_h - x_f - hd_1) + k_i(x_l - x_f - hd_2)}{(C_{air}.v_f + 2k_d)} \right] \quad (5)$$

The desired distance between the header and the follower vehicle and between the leader and the follower vehicle are denoted as hd_1 and hd_2 respectively. The gain parameters k_p, k_i and k_d have usual interpretation.

III. Platoon Maneuvers

Most of the present-day researchers in [2] [3] [4] have focused on their studies on platooning following the basic assumption that the platoons start from the base station and remain as such throughout the venture. This paper tries to improve this used case scenario circumscribing the most basic platoon management scenarios such as the split and the merge. While in the split scenario where one platoon (with at-least 2 vehicles) breaks up into two smaller platoons and travels to their defined intended locations, two platoons travelling in a same lane merges to become one single platoon in the merge scenario.

Figure 2 vividly depicts the merging scenario of two platoons with 3 and 4 HDVs respectively in them. The header vehicle of each platoon has been color coded to red while the rest of the platoon members are coded to be black for the sake of clarity.

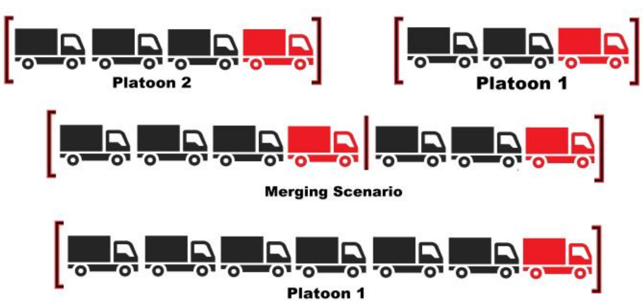

Figure 2: Platoon Merge Scenario

The merge maneuver is always commenced by the header of the platoon 2 when the platoons come close to each other within some prescribed distance. The distance is generally judged based on the communication constraint between the vehicles and has not been described here as it is not within the scope of the work. The merging scenario can be realized in three different steps: Appeal, Reply and Implementation (abbreviated as A.R.I) as in the Figure 3.

Figure 3: ARI protocol for platoon maneuvers.

Appeal$_{Merge}$: The header of the platoon 2 as shown in Figure 2 appeals for platoon merging to the header of the first platoon, i.e. platoon 1. This communication is generally realized wirelessly and is achieved using the Dedicated Short Range Communication (DSRC) which has been standardized in IEEE 802.11. There are several reasons that DSRC has been used. The most prominent reason has been the fact that DSRC operates at around 5.9 GHz. It is designed to support various operations and has a high transmission range varying from 100-1000 meters which is highly suitable for data-exchange between vehicles in the platoons. It has to be borne in mind that there can be certain cases of the appeal not raised due to communication failure between the HDVs, and in such cases the appeal requests from the trailing platoon which wishes to join the platoon ahead would necessarily has to be re-raised.

Reply$_{Merge}$: The header of the platoon 1 either accepts or rejects the appeal based on its current scenario. Reasons for rejection could be such that the platoon 1 could be going through some other complex platoon management scenarios themselves such as lane changing operations of its own or may be due to a communication failure. In such cases, platoon 2 needs to wait until a valid acknowledgement is received.

Implementation$_{Merge}$: Once the acknowledgement has been received and that the merging scenario has been permitted, there could be a proper information interchange between the last vehicle of the platoon 1 and the header of the platoon 2. The data packet which has the platoon 2's label addresses are exchanged between the vehicles of platoon 1 and vice-versa. The speed of the platoon 1 and platoon 2 have been matched and the control action starts considering the distance between the header of the platoon 2 and the last follower of platoon 1 as error signals. This entire protocol takes place in a fraction of seconds' depending on the communication interface between the platoons. The control action decreases the intervehicular gap of the platoons and matches the inter-vehicular gap hd_1 and hd_2 as mentioned in the earlier section. Once the merging scenario has been realized, the control of the entire merged platoon shifts directly to the header of the platoon 1 and the entire set of platoon of vehicles move as intended. The split scenario can be similarly realized for the platoons. The only difference has been the implementation section which has been explained below. The ***Appeal$_{Split}$*** and ***Reply$_{Split}$*** have been omitted in this section for the sake of brevity.

Implementation$_{Split}$: As similar to the platoon merging scenario, the split scenario has also been realized. From the Figure 4 it can be seen that the third vehicle from platoon 1 has put forward a split request. Once the split request from the third vehicle has been acknowledged by the header of the first platoon, the third vehicle is assigned to be the header of platoon 2. The speed of the platoon 2 is reduced and the platoon 2 naturally falls back in distance from platoon 1, realizing the split. Once the platoons are split, the platoon 2 is dealt as a standalone platoon and can move as desired.

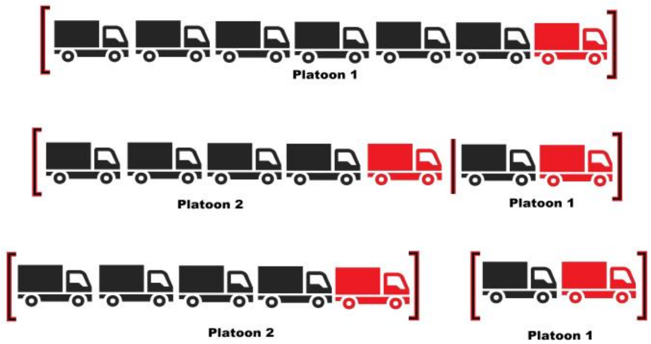

Figure 4: Platoon Split Scenario

IV. Simulation and Results

In this section, the platoon maneuver scenario has been realized using MATLAB and VISSIM [16]. VISSIM is a micro-simulator which has the capabilities to replicate and control the trajectories of vehicles in simulation for both right hand and left hand driving scenarios. This is one of the major advantages of using VISSIM over the open source traffic simulator SUMO [17] which allows only left hand traffic driving scenarios. VISSIM even allows the access of vehicular information from infrastructure as well as from other vehicles with simulation time as low as 100ms.


Figure 5: The road section of Alexandria Road, Singapore

With simulation time as low as this, it is generally easy to track the vehicle parameters to the smallest time interval, thus providing control on the entire platoon of HDVs. Figure 5 shows the Alexandria Road freeway corridor in Singapore as implemented using the "Toggle Background Maps" function in VISSIM. The function has been used to overlay the required section of the road on the maps. The vehicles have been injected into the network at the exact link and the lane as desired by the user. VISSIM allows certain types of vehicle models, out of which the HDVs were used for the same as the controller has been developed for platooning of trucks in a road network. Considering the challenges of extending the simulation to a much more holistic environment, the controller algorithm uses the "*Co-ord*" function. The co-ordinate function refers to the exact GPS position of the trucks with respect to the entire map area plotted using VISSIM. The advantage of using the GPS coordinates come from the fact that the controller code for the algorithm can be extended to any road networks by just putting the layover map of the desired location. The simulation has been done for a platoon of 14 HDVs moving from a base station to another while moving across a freeway. The speed profile of the HDVs are shown in Figure 6 with a total simulation period of 1000 seconds. The controller algorithm used in (5) has been used for a set of 14 vehicle platoon with gain parameters as $k_p = 900$, $k_i = 2250$ and $k_d = 200$. It has to be borne in mind that the controller gains above used for this simulation study has been extracted using the classical control methodologies. The methodologies have not been elucidated in here as it is outside of the scope of this present work. Figure 6 shows the speed tracking of the header of the platoon by the rest of the platoon vehicles. While the header starts at a speed of 30 km/h, it increases speed to 50 km/h, and ultimately to 70 km/h. The speed stays at 70 km/h for a certain time until it reduces to 51 km/h. Figure 6 shows the close tracking of the platoon followers to the platoon header while the speed profile varies within the simulation proving the efficacy of the controller.

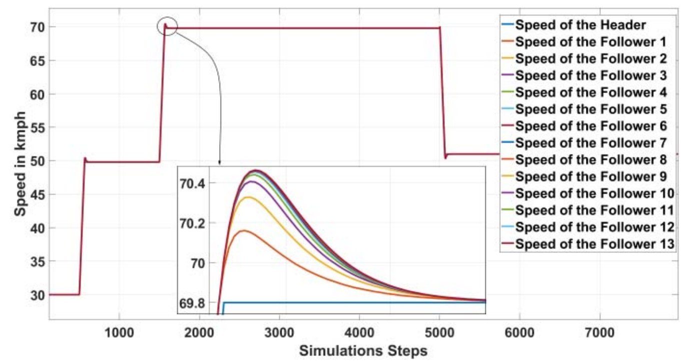

Figure 6: Speed profile of the 14 vehicle platoon members

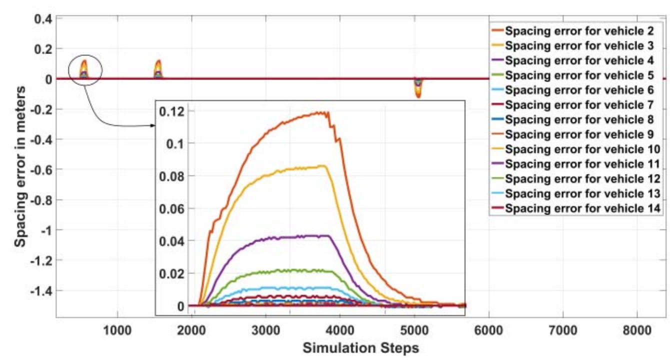

Figure 7: Distance profile of the platoon members

The spacing error in distance of each vehicle in the platoon has been shown in Figure 7 above. It can be seen that the proposed controller can keep the spacing error between the vehicles to be within ± 0.12 meters.

The next section illustrates the platoon maneuvers using the same controller.

(i) ***Merge Maneuver:*** For the merge maneuver, a platoon length of 14 HDVs have been considered. The two platoons move at a fixed speed with **$GAP_{DESIRED}$** between them. The work done in this section follows an assumption that last follower of the first platoon and the header of the second platoon cannot be more than the maximum allowable gap (**GAP_{MAX}**) while the ***$Appeal_{Merge}$*** has been raised. On a highway when the platoons are ready for a merge, they need to be close to each other for the ***Appeal***$_{Merge}$ to be raised and accepted. In-case, if there are any random intervehicular gap between the platoons (more than **GAP**$_{MAX}$), the communication between the vehicles would fail and hence the merging maneuver will not be realized. The prescribed distance for the merge hasn't been described here, as the communication scenario and its limits are not within the scope of the present work. For the simulation scenario, it is considered that first platoon move at 50 km/h at while the second moves at 40 km/h. After a certain time when the header of platoon 2 follows the **A.R.I** protocol as mentioned above and raises an ***Appeal***$_{Merge}$ to join, the header of the platoon 1 accepts the ***Requests***$_{Merge}$ and implements the same for the set of vehicles in the platoon.

It is also to be seen here that the merging of the vehicles in the platoon strictly follows the specific speed limits of the concerned road networks of the country. For example, as the maximum speed of HDVs in Singapore is 70 km/h the simulation is in accord with this speed protocol. Figure 8 below shows the speed curve for the set of platoons that has been involved in the merge scenario. The curve in the Figure 8 shows platoons 1 move at 50 km/h while platoon 2 moves at 40 km/h. From the graphs, it can be seen that after a certain time, the merge scenario has been initiated by the platoon 2. The platoon 2 which has been lagging behind in distance from the platoon 1 has to increase speed in accordance with the controller algorithm mentioned in (5) and hence is evident from the curves.

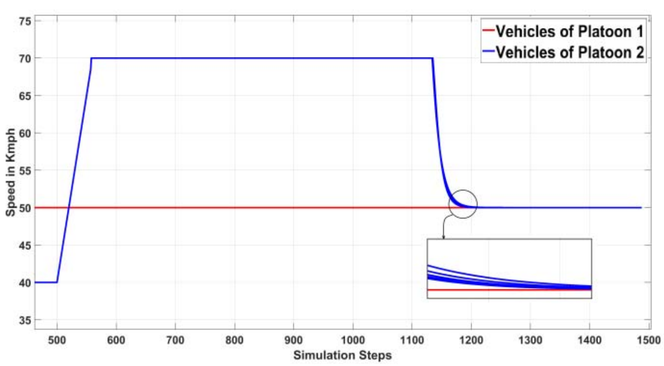

Figure 8: Speed profile of the platoon members

It can also be seen that the speed of the vehicles in platoon 2 has increased to 70 km/h to ultimately close the gap between the platoons to the desired distance. The speed of platoon 1 is at constant 50 km/h. The speed of the platoon 2 has been within the acceptable limit and the time of the entire merging scenario could be seen to take place within 700 simulation steps (between 500 and 1200), which is equivalent to 70 seconds in real time (resolution is 0.1). Figure 9 shows the intervehicular distance between the platoon members while merging. The curve represents the distance gap between the two individual platoons before the merge scenario has been realized. It can be seen that the distance is around 100 meters between the last vehicle of first platoon and the header of the second platoon before the merge scenario has been realized.

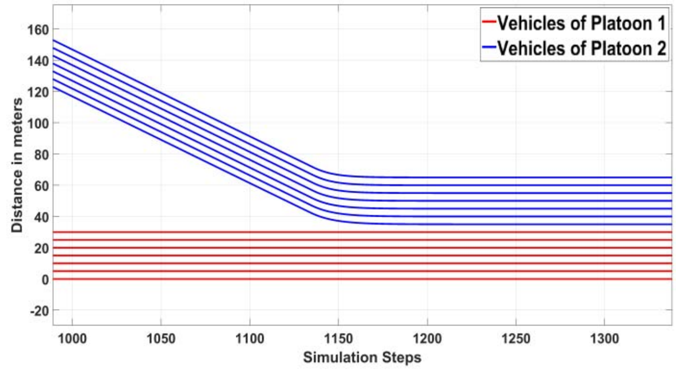

Figure 9: Intervehicular distance between vehicles of platoon 1 and 2

Once the merge scenario has been realized, the header of the second platoon increases its speed as shown in Figure 8 and closes in the gap with the last vehicle of platoon 1. It can be clearly seen that the **GAP**$_{DESIRED}$ of 5 meters have been achieved and the entire merged platoon of 14 vehicles move together.

(ii) ***Split Maneuver****:* The split maneuver has been realized in a different methodology. While a platoon of 14 vehicles are seen to move in tandem, it has been considered that the two platoons must split after the 7th vehicle. Once the request has been raised and acknowledged, the vehicle 8 is assigned to be the header of platoon 2 as mentioned in section III above. Figure 10 below shows the speeds of all the vehicles in the platoon to be around 70 km/h before the commencement of the split maneuver. Once the ***Appeal***$_{Split}$ and ***Reply***$_{Split}$ have been accepted by the header of the 14 vehicle platoon, the ***Implementation***$_{Split}$ starts by decreasing the speed of the platoon 2. The decrease in speed can be seen from the Figure 10 where the vehicles speeds wane to 51 km/h. With the reduced speed, the distance between the platoons increase until the two platoons are dealt as standalone ones. The increase in intervehicular distance between the last vehicle of the platoon 1 and the vehicles of platoon 2 after the split is also evident from the Figure 11. While the position of the platoon 1 has been kept as reference, the ever increasing distance of the other vehicles in platoon 2 clearly depicts that the split maneuver has been properly realized

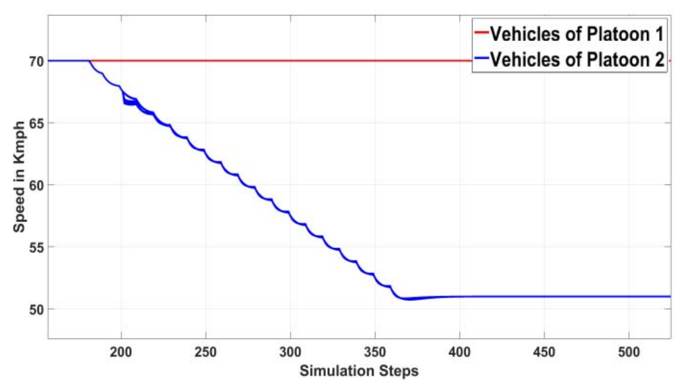

Figure 10: Speed profile of platoon 1 and platoon 2 while splitting

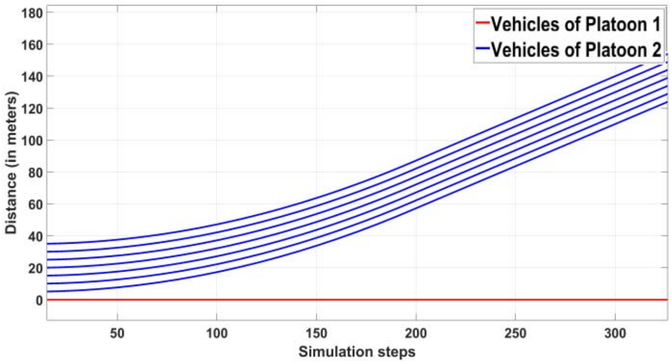

Figure 11: Intervehicular distance between the reference vehicle and 8 vehicles of platoon 2

V. Comparison With Existing Methodology

This section of the paper compares the results obtained from this work with the existing controller methodology in Amoozadeh *et.al.* The controller developed in [18] has been used on HDVs traversing on the same road profile as shown in Figure 5. From Figure 12, it can be seen that using the controller in [18], there has been a prominent undershoot in the speed curve when the merging protocol commence which differs from the speed profile in Figure 8. It can also be seen that using the controller in [18], the merge protocol has been completely realized within 900 simulation steps approximately which is around 90 seconds, 20 seconds more than one shown in Figure 8.

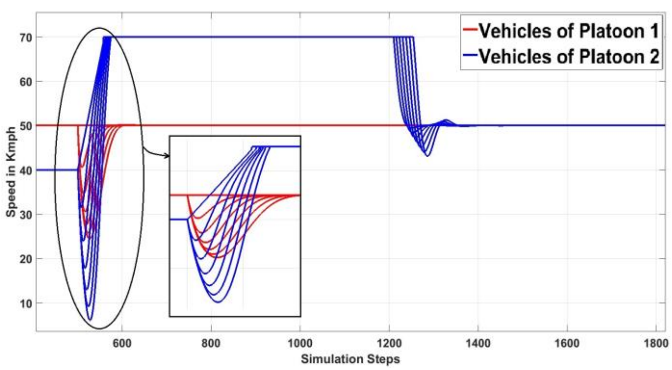

Figure 12: Speed profile of the platoon while merging using the controller in [18].

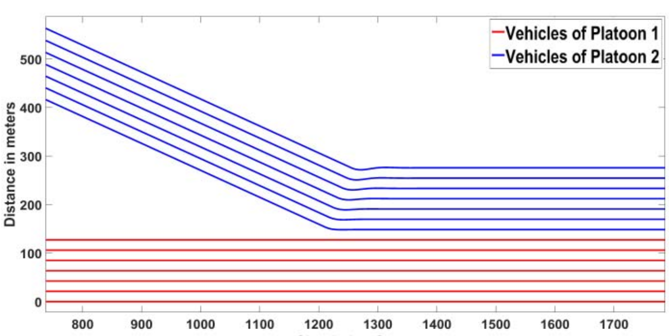

Figure 13: Intervehicular distance between vehicles of platoon 1 and 2 using the controller in [18] while merging.

Similarly for the splitting maneuver, the speed and inter-vehicular distance curves in Figures 14 and 15 are baldly different from the ones in Figures 10 and 11 proving the efficacy of the controller in (5). The speed profile shows undershoot when the splitting maneuver has been commenced. Using the same controller in [18], the intervehicular distance has been around 12 meters which should have matched the $\mathbf{GAP_{DESIRED}}$ of 5 meters.

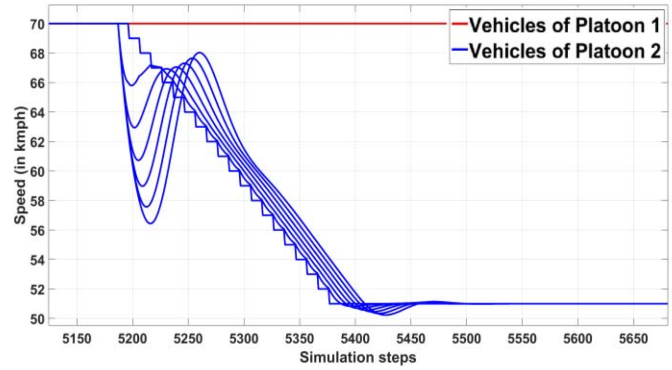

Figure 14: Speed profile of the platoon while splitting using the controller in [18].

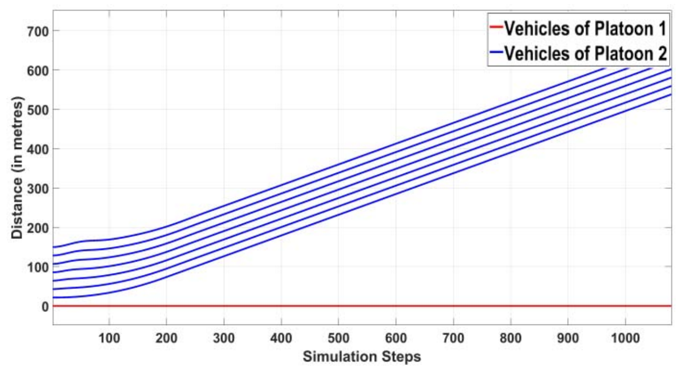

Figure 15: Intervehicular distance between vehicles of platoon 1 and 2 using the controller in [18] while splitting.

Conclusion and Future Work

In this paper, two platoon maneuvers have been simulated, i.e., the splitting and the merging maneuver. While platoon maneuvers have been attempted by a number of researchers, this paper proposes a PID controller methodology to work out the splitting and merging scenario of platoons. The result section shows the efficacy of the controller not only for the platoon movement, but also for the platoon maneuvers as well. The curves illustrates that the platoon of vehicles is string stable. The string stability in a vehicle platoon guarantees that any disturbance that appears in between the vehicles does not propagate to the end of the platoon stream thereby making the entire system to be stable in terms of coercive movements. While in this case, it has been considered that the platoon vehicles communicate between themselves, the future work would introduce V2V communication between the non-platoon members as well with much more complex communication scenario for analysis. The split and merge maneuvers in such scenarios would also be more challenging as there would be multiple data-drops hindering the smooth realization of such maneuvers. The implementation of the PID controller for splitting and merging of platoons using Hardware In-Loop (HIL) is also in order.

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